

# Database modelling

*MBUA 512*

# Class 2 – where we're going

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*Database modelling, with Michael Winikoff*

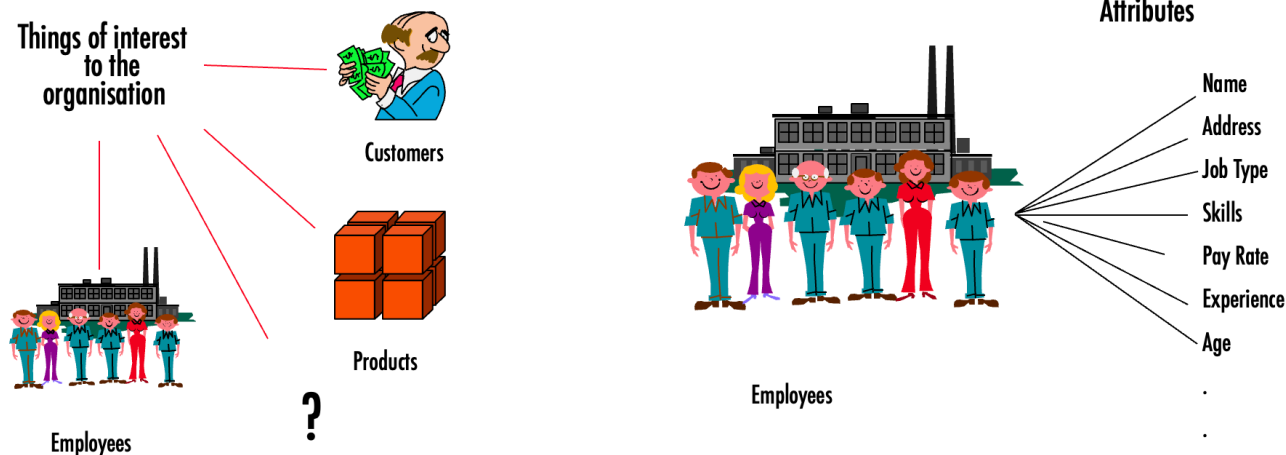
1. Data modelling using diagrams
2. Translating diagrams to relational databases, and the role of *keys*
3. Anomalies, and avoiding them by *normalisation*

*Based on slides by Professor Markus Luczak-Roesch.*

# Recap from last week

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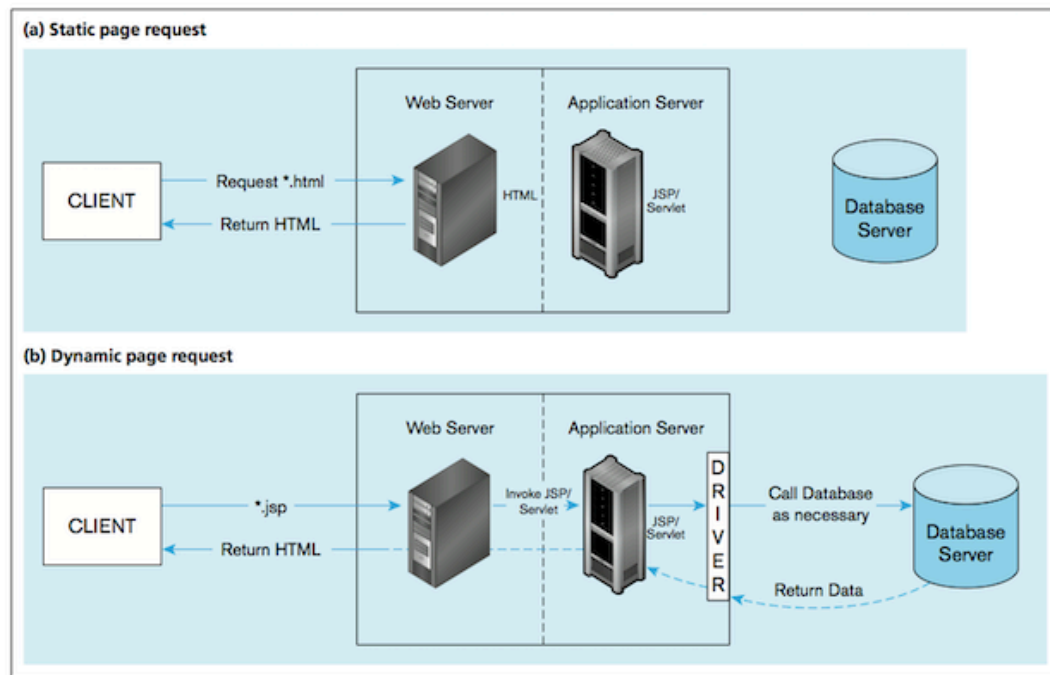
- What is data?
  - Types of data (qualitative/quantitative, structured/unstructured)
  - Why is data important ... to organisations? ... to business analysts?
- What is a DB management system and what is a *relational* database? (technology)
- Concepts: **entity**, **attributes**, **relationships** (modelling data)

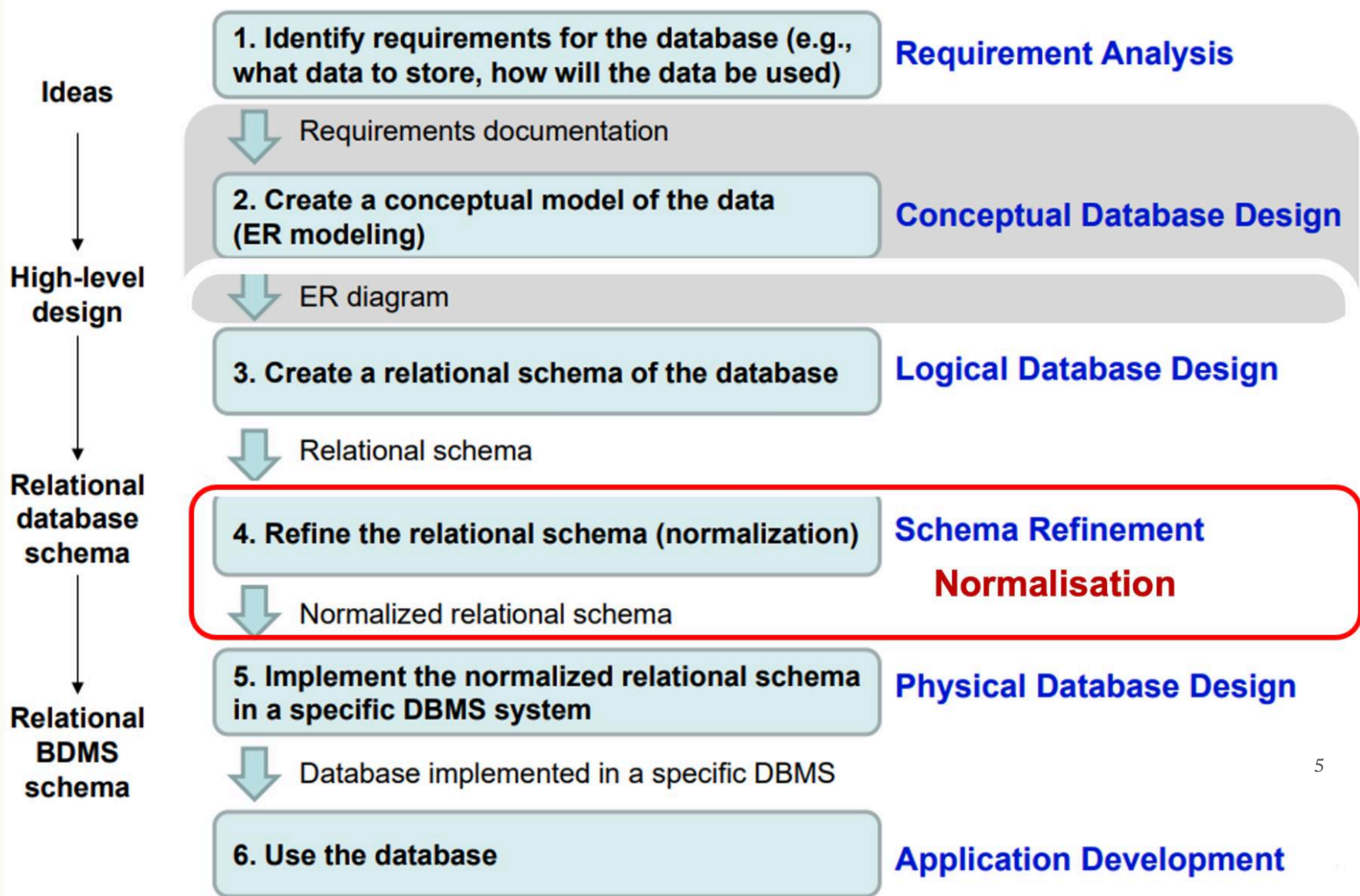


# Big picture

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**Roles:** DB analysis, designers, developers, front-end app analysts and developers, DB admin





# Data modelling using diagrams

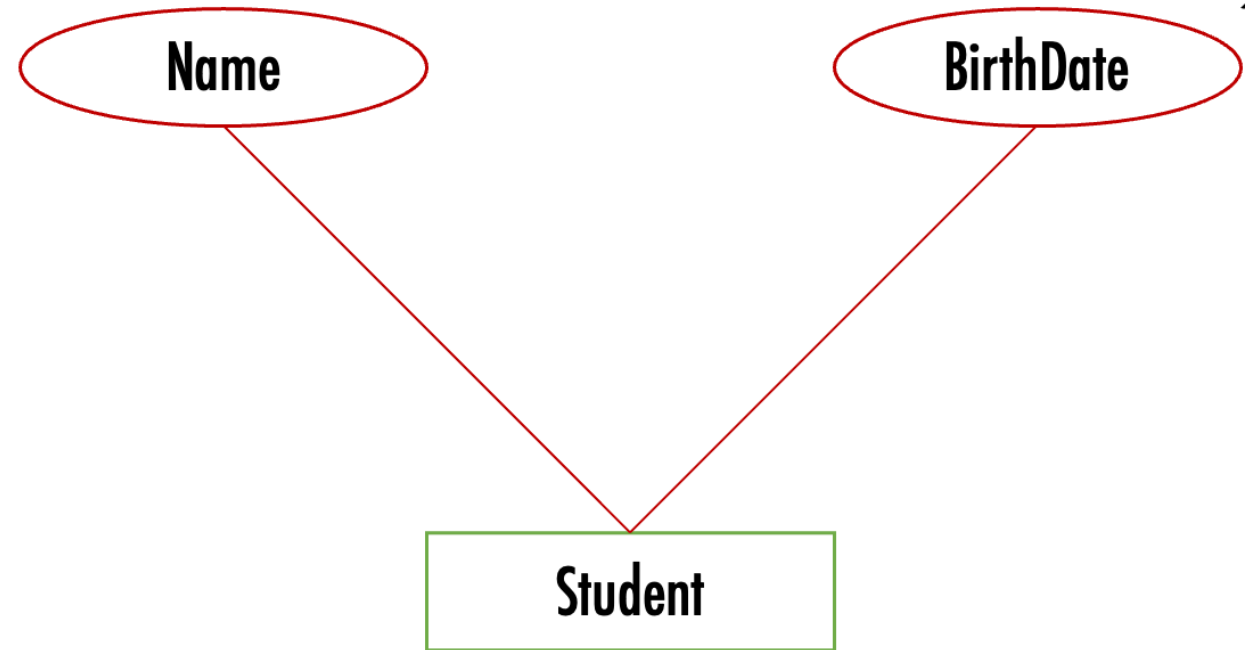
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# 1. Data modelling using diagrams

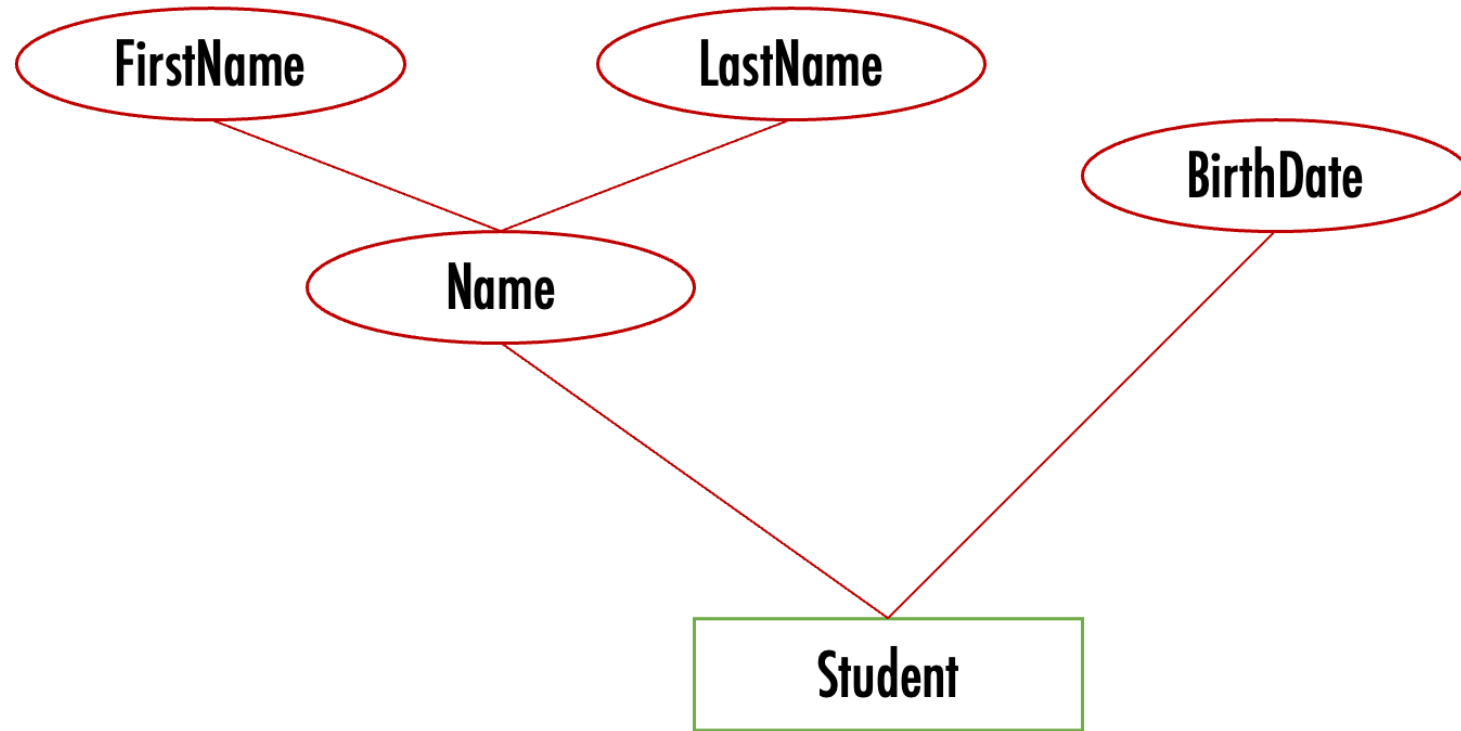
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Model *conceptual* model in terms of entities, attributes, relationships using an **ER (Entity-Relationship) diagram**:

- Entity = rectangle
- Attribute = oval
- Relationship = Diamond

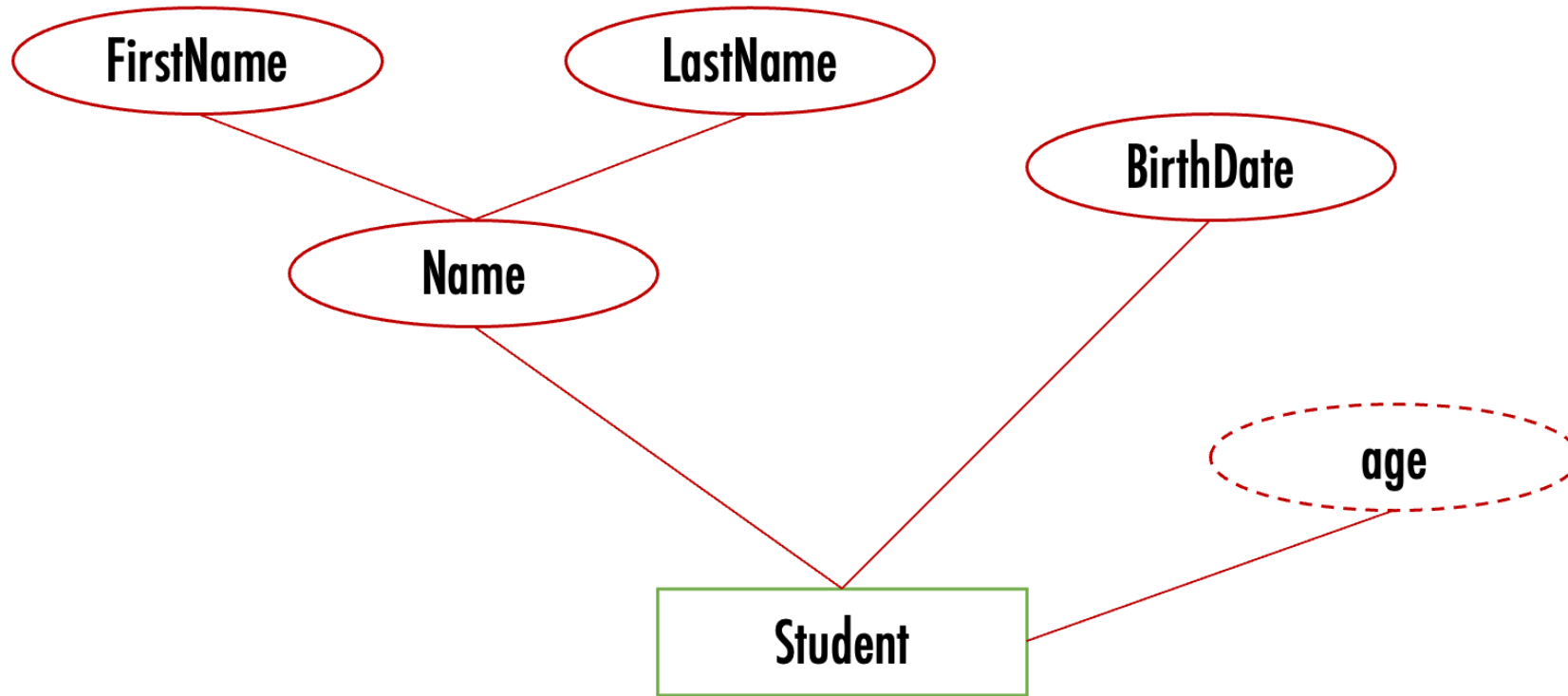


Attributes can be *composite*

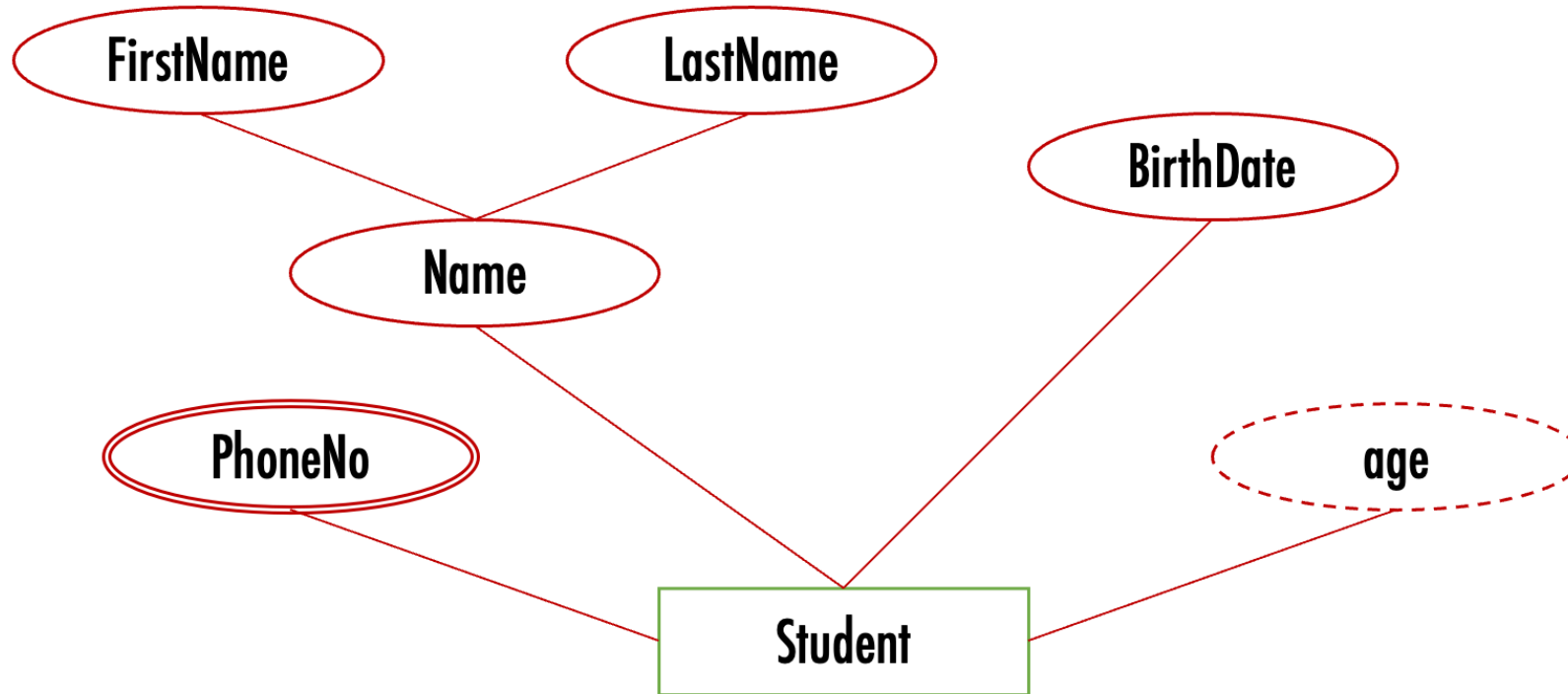




Attributes can be *derived* (dashed oval)

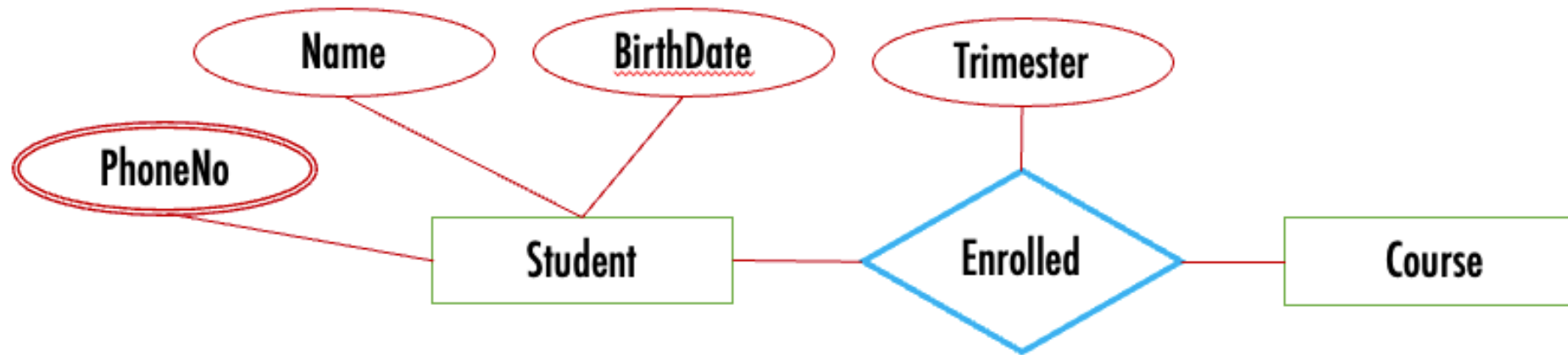


Attributes can have *multiple values* (double edged oval)



## Relationships ...

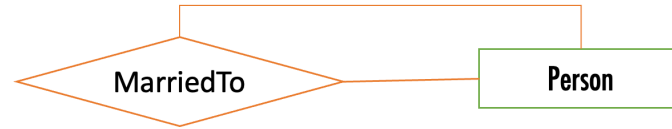
- A relationship links one or more entities, and can have attributes



- Relationships have **degree** and **cardinality**:
  - Degree: number of participating entity **types** in a relationship
  - Cardinality: number of entity **instances**

# Degree – number of types of entities

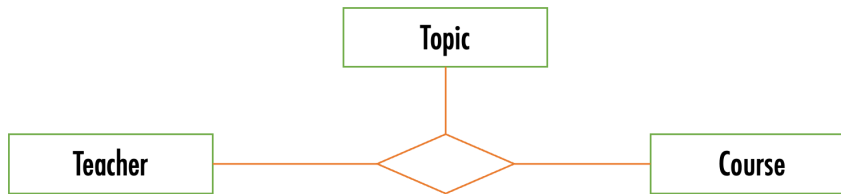
- Unary



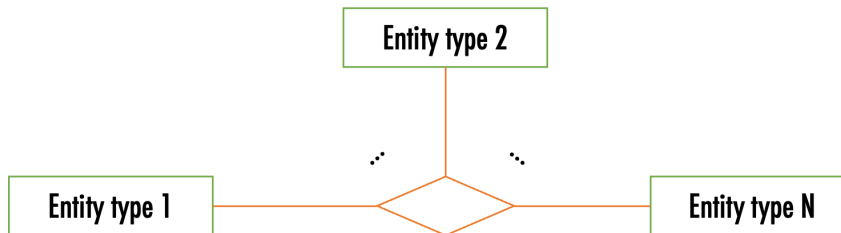
- Binary



- Ternary

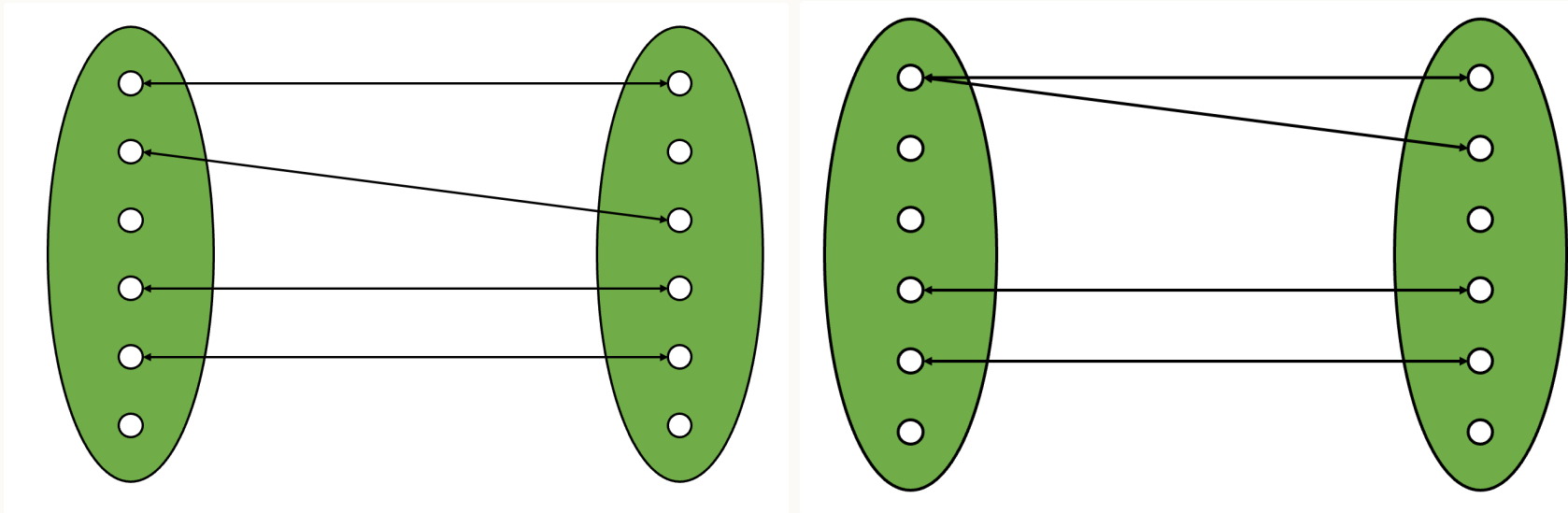


- n-ary



## Cardinality – number of *instances* of entities

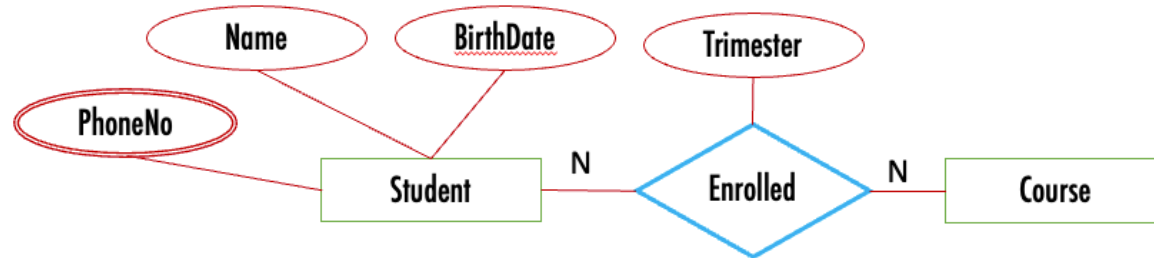
Left: one-to-one (e.g. bus and bus driver); Right: one-to-many (e.g. bus and passenger)



## Examples of degree and cardinality

- MarriedTo is Unary degree: only one type of entity is involved: a Person
- MarriedTo is 1-1 cardinality: each individual person is married to one other person
- Teaches is Binary degree: two entity types, teacher and course
- Teaches is N-N cardinality: a teacher can teach many courses, and a course can have many teachers

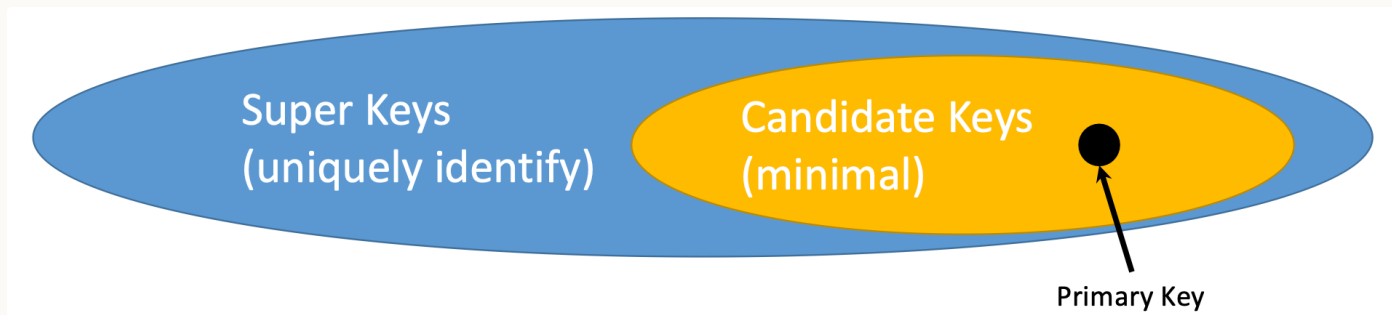
# Notation for cardinality



# Keys

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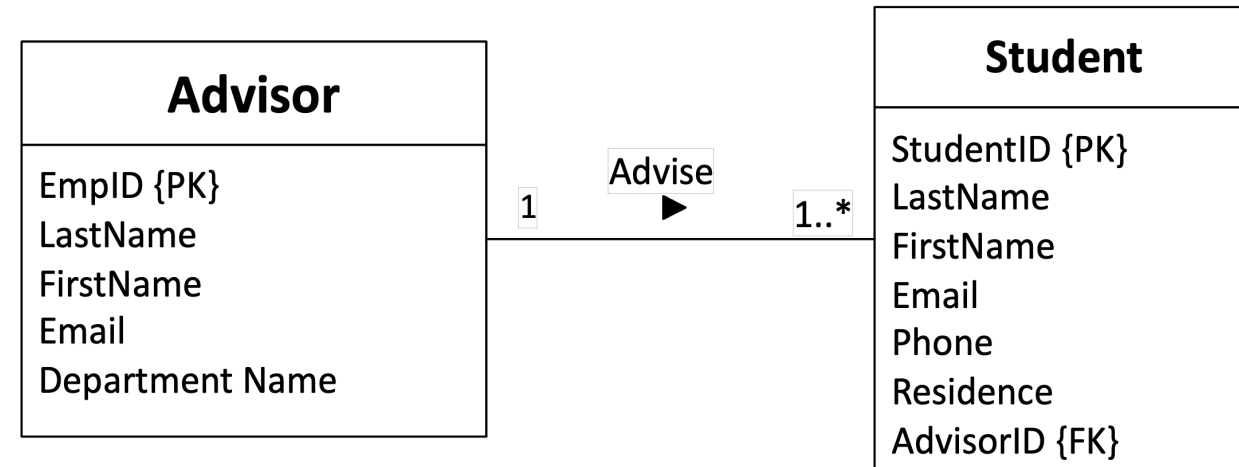
- A **key** is one or more attributes that must be unique to each entity instance, i.e. no two instances can have the same value for their key attribute(s)
  - Example: name is a bad choice for key - two people can have the same name
- A *super key* is a set of attributes that can be a key
- A *candidate key* is a *minimal* super key (no subset of it is a super key)
- A *primary key* is a chosen candidate key
- Notation: key attribute is underlined





# Alternative notation: UML (Unified Modeling Language)

- Rectangle = Entity, with attributes listed in the rectangle
- Relationship = line, with cardinalities indicated using annotation at each end (e.g. "1", "0..", "1.." - "\*" means "many")
- Keys indicated with {PK}
- Relationships with attributes indicated with a dashed line from the relationship to a rectangle (not shown here)



# Tips for modelling

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- Iterative!
- Need to focus on what is important, rather than capture everything
- Careful with notation: link entities and associations with their attributes, an association needs to be linked to one or more entities
- Naming:
  - Brief, but clear
  - Unique - e.g. don't use "name" for both "student name" and "employee name"
  - Entities and attributes: singular ("student")
  - Relationships: verbs or verb phrases ("inspects", "manages", "belongsTo")

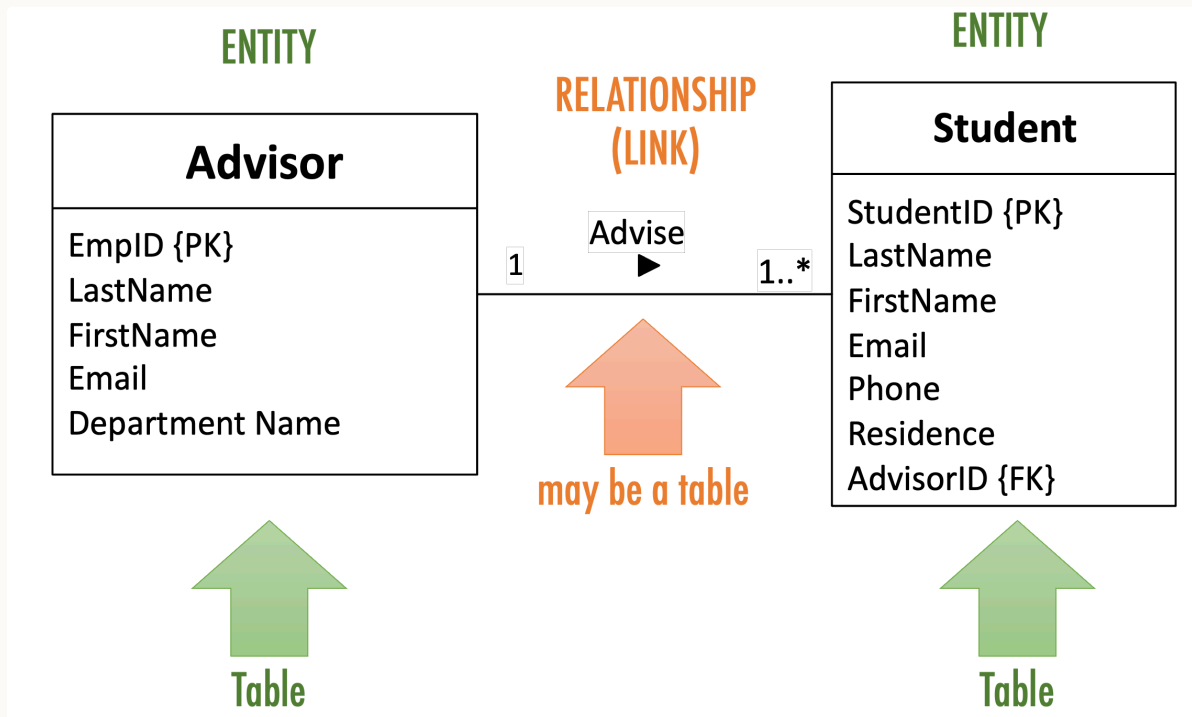
# Translating diagrams to relational databases

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## 2. Translating diagrams to relational databases

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- Each entity becomes a table (relation) containing the entity's attributes
- Each relationship is mapped to either a table or additional attributes, depending on the cardinalities



## Mapping a 1-N relationship as an additional attribute



| EmpID | LastName |
|-------|----------|
| 2463  |          |
| 1942  |          |
|       |          |

| StudentID | LastName | Advisor |
|-----------|----------|---------|
| 4511      | ...      | 2463    |
| 8463      | ...      | 2463    |
| 9617      | ...      | 1942    |

# Mapping an N-N relationship as a table



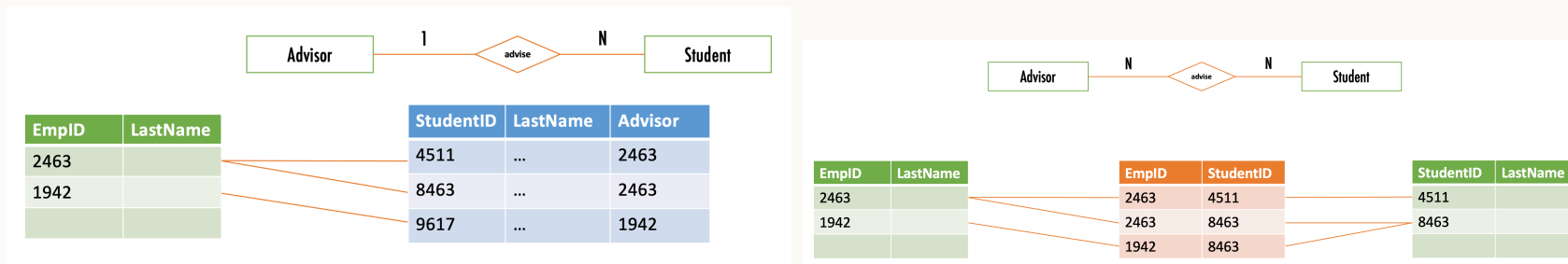
| EmpID | LastName |
|-------|----------|
| 2463  |          |
| 1942  |          |
|       |          |

| EmpID | StudentID |
|-------|-----------|
| 2463  | 4511      |
| 2463  | 8463      |
| 1942  | 8463      |

| StudentID | LastName |
|-----------|----------|
| 4511      |          |
| 8463      |          |
|           |          |

# Foreign keys

- A *key* is the attribute(s) that can be used to identify an entity instance (e.g. StudentID)
- Each table (relation) will have column(s) for its keys (e.g. the Student table has a column StudentID)
- A *foreign key* is when a table has a column that contains keys for *another table* (e.g. Student table contains the ID of the student's advisor)
  - in UML notation indicated with {FK}



# Anomalies and Normalisation

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# 3. Anomalies and Normalisation

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Errors or inconsistencies occurring when updating tables with redundant data:

- insertion anomaly example: require data about different entities to be entered at the same time in a single relation – cannot track customer address separately from order
- deletion anomaly example: deletion of a row results in loss of multiple facts – deleting the last order loses the customer's contact details!
- modification anomaly example: need to modify multiple rows to update a fact consistently – update address requires updating every order

Example on the next slides.

AD CAMPAIGN MIX

| <u>AdCampaignID</u> | AdCampaign Name | StartDate | Duration | Campaign MgrID | Campaign MgrName | <u>ModusID</u> | Media | Range    | Budget Pctg |
|---------------------|-----------------|-----------|----------|----------------|------------------|----------------|-------|----------|-------------|
| 111                 | SummerFun20     | 6.6.2020  | 12 days  | CM100          | Roberta          | 1              | TV    | Local    | 50%         |
| 111                 | SummerFun20     | 6.6.2020  | 12 days  | CM100          | Roberta          | 2              | TV    | National | 50%         |
| 222                 | SummerZing20    | 6.8.2020  | 30 days  | CM101          | Sue              | 1              | TV    | Local    | 60%         |
| 222                 | SummerZing20    | 6.8.2020  | 30 days  | CM101          | Sue              | 3              | Radio | Local    | 30%         |
| 222                 | SummerZing20    | 6.8.2020  | 30 days  | CM101          | Sue              | 5              | Print | Local    | 10%         |
| 333                 | FallBall20      | 6.9.2020  | 12 days  | CM102          | John             | 3              | Radio | Local    | 80%         |
| 333                 | FallBall20      | 6.9.2020  | 12 days  | CM102          | John             | 4              | Radio | National | 20%         |
| 444                 | AutmnStyle20    | 6.9.2020  | 5 days   | CM103          | Nancy            | 6              | Print | National | 100%        |
| 555                 | AutmnColors20   | 6.9.2020  | 3 days   | CM100          | Roberta          | 3              | Radio | Local    | 100%        |

## AD CAMPAIGN MIX

| <u>AdCampaignID</u> | <u>AdCampaign Name</u> | <u>StartDate</u> | <u>Duration</u> | <u>Campaign MgrID</u> | <u>Campaign MgrName</u> | <u>ModusID</u> | <u>Media</u> | <u>Range</u> | <u>Budget Pctg</u> |
|---------------------|------------------------|------------------|-----------------|-----------------------|-------------------------|----------------|--------------|--------------|--------------------|
| 111                 | SummerFun20            | 6.6.2020         | 12 days         | CM100                 | Roberta                 | 1              | TV           | Local        | 50%                |
| 111                 | SummerFun20            | 6.6.2020         | 12 days         | CM100                 | Roberta                 | 2              | TV           | National     | 50%                |
| 222                 | SummerZing20           | 6.8.2020         | 30 days         | CM101                 | Sue                     | 1              | TV           | Local        | 60%                |
| 222                 | SummerZing20           | 6.8.2020         | 30 days         | CM101                 | Sue                     | 3              | Radio        | Local        | 30%                |
| 222                 | SummerZing20           | 6.8.2020         | 30 days         | CM101                 | Sue                     | 5              | Print        | Local        | 10%                |
| 333                 | FallBall20             | 6.9.2020         | 12 days         | CM102                 | John                    | 3              | Radio        | Local        | 80%                |
| 333                 | FallBall20             | 6.9.2020         | 12 days         | CM102                 | John                    | 4              | Radio        | National     | 20%                |
| 444                 | AutmnStyle20           | 6.9.2020         | 5 days          | CM103                 | Nancy                   | 6              | Print        | National     | 100%               |
| 555                 | AutmnColors20          | 6.9.2020         | 3 days          | CM100                 | Roberta                 | 3              | Radio        | Local        | 100%               |

The media and range values for campaign modus 1 repeated twice

The name of the campaign manager CM100 repeated three times

The name, start date, and duration of the campaign 222 repeated three times

AD CAMPAIGN MIX

| <u>AdCampaignID</u> | AdCampaign Name | StartDate | Duration | Campaign MgrID | Campaign MgrName | <u>ModusID</u> | Media    | Range    | Budget Pctg |
|---------------------|-----------------|-----------|----------|----------------|------------------|----------------|----------|----------|-------------|
| 111                 | SummerFun20     | 6.6.2020  | 12 days  | CM100          | Roberta          | 1              | TV       | Local    | 50%         |
| 111                 | SummerFun20     | 6.6.2020  | 12 days  | CM100          | Roberta          | 2              | TV       | National | 50%         |
| 222                 | SummerZing20    | 6.8.2020  | 30 days  | CM101          | Sue              | 1              | TV       | Local    | 60%         |
| 222                 | SummerZing20    | 6.8.2020  | 30 days  | CM101          | Sue              | 3              | Radio    | Local    | 30%         |
| 222                 | SummerZing20    | 6.8.2020  | 30 days  | CM101          | Sue              | 5              | Print    | Local    | 10%         |
| 333                 | FallBall20      | 6.9.2020  | 12 days  | CM102          | John             | 3              | Radio    | Local    | 80%         |
| 333                 | FallBall20      | 6.9.2020  | 12 days  | CM102          | John             | 4              | Radio    | National | 20%         |
| 444                 | AutmnStyle20    | 6.9.2020  | 5 days   | CM103          | Nancy            | 6              | Print    | National | 100%        |
| 555                 | AutmnColors20   | 6.9.2020  | 3 days   | CM100          | Roberta          | 3              | Radio    | Local    | 100%        |
| ????                | ????            | ????      | ????     | ????           | ????             | 7              | Internet | National | ????        |

**Deletion Anomaly Example:**  
Cannot delete campaign 444 without also deleting all the data about campaign manager CM103 and campaign modus 6.

**Modification Anomaly Example:**  
To change the duration of the campaign 222 from 30 to 45 days, three records have to be modified.

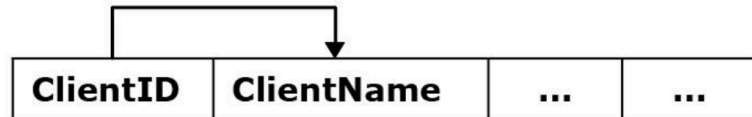
**Insertion Anomaly Example:**  
Cannot insert new campaign modus 7 without inserting an actual campaign using the new modus 7.

**Solution:** *normalise* by splitting up the table

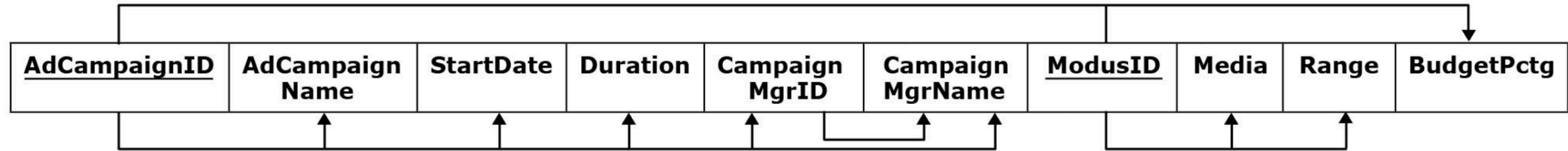
**How to do this?** By analysing *functional dependencies*.

**Functional dependency** - occurs when the value of one (or more) column(s) in each record of a relation uniquely determines the value of another column in that same record of the relation

Example:  $\text{ClientID} \rightarrow \text{ClientName}$

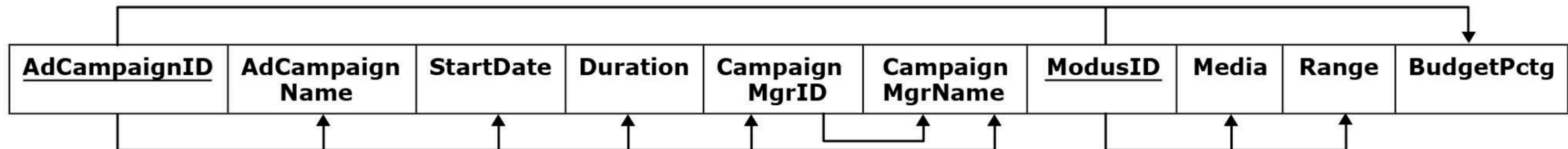


## Dependencies for the ad campaign DB

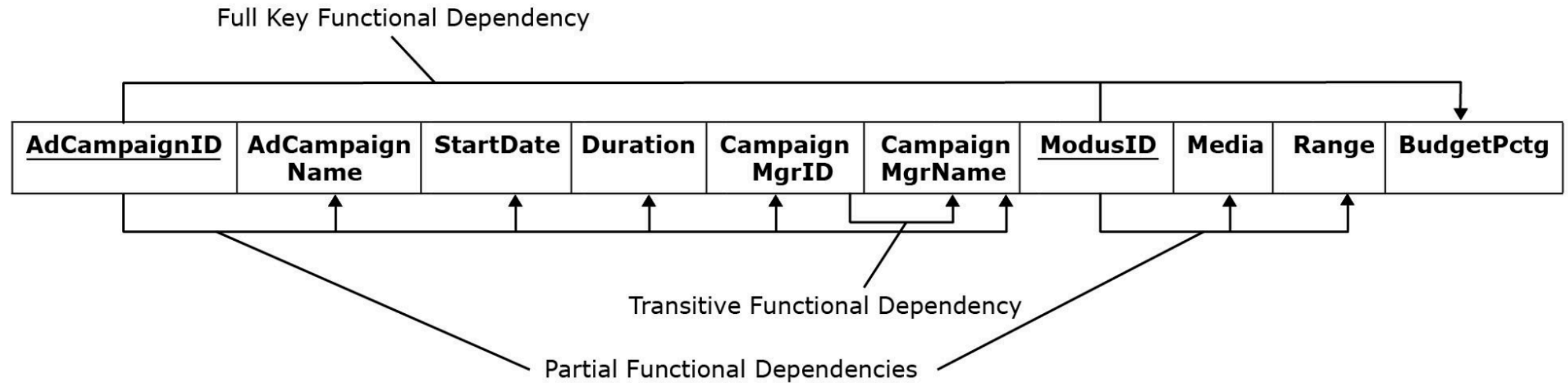


# Types of functional dependencies

- **✗ Partial:** a column depends on a *component* of a composite primary key; e.g.  
ModusID → Media, Range
- **✓ Full:** the primary key functionally determines the column, and no separate component of it partially determines the same column
- **✗ Transitive:** nonkey columns functionally determine other nonkey columns; e.g.  
CampaignMgrID → CampaignMgrName



# Dependencies for the ad campaign DB





# Normalisation - 1NF

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- First Normal Form (1NF): no multi-value entries
- Normalise by splitting records

VET CLINIC CLIENT

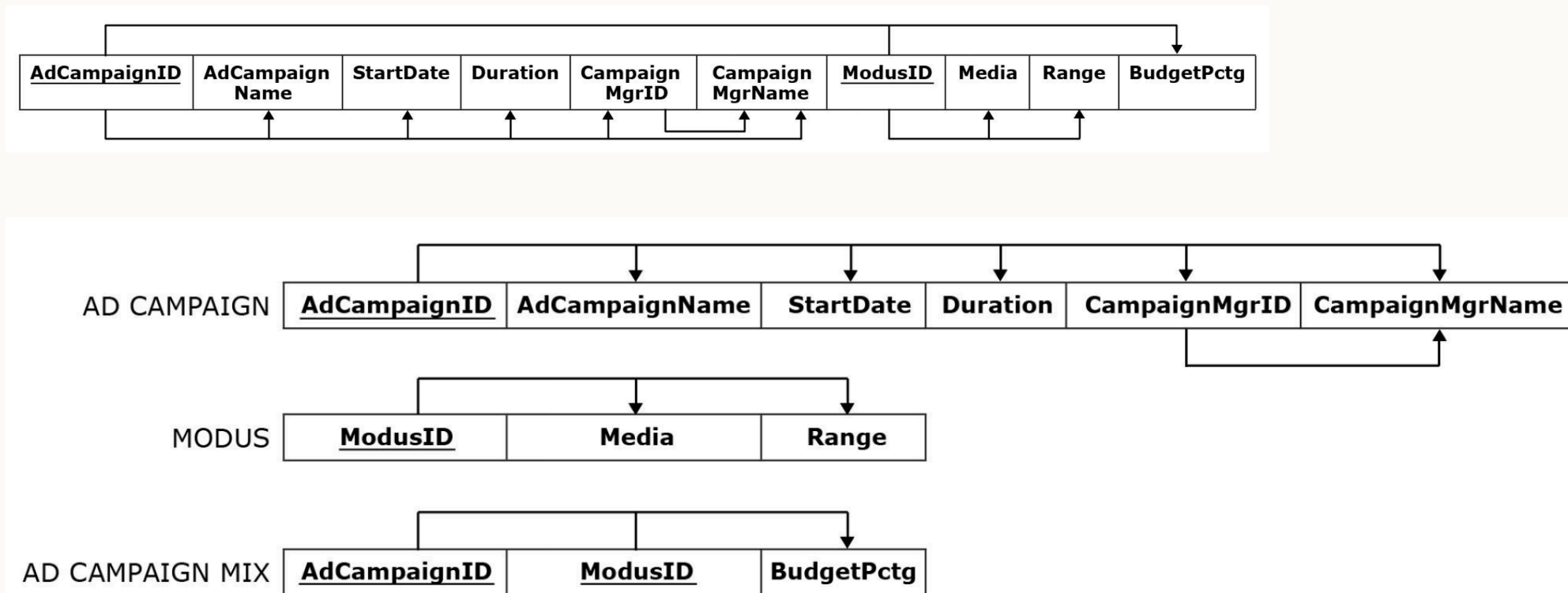
| <u>ClientID</u> | ClientName | PetNo | PetName | PetType |
|-----------------|------------|-------|---------|---------|
| 111             | Lisa       | 1     | Tofu    | Dog     |
| 222             | Lydia      | 1     | Fluffy  | Dog     |
|                 |            | 2     | JoJo    | Bird    |
|                 |            | 3     | Ziggy   | Snake   |
| 333             | Jane       | 1     | Fluffy  | Cat     |
|                 |            | 2     | Cleo    | Cat     |

VET CLINIC CLIENT

| <u>ClientID</u> | ClientName | <u>PetNo</u> | PetName | PetType |
|-----------------|------------|--------------|---------|---------|
| 111             | Lisa       | 1            | Tofu    | Dog     |
| 222             | Lydia      | 1            | Fluffy  | Dog     |
| 222             | Lydia      | 2            | JoJo    | Bird    |
| 222             | Lydia      | 3            | Ziggy   | Snake   |
| 333             | Jane       | 1            | Fluffy  | Cat     |
| 333             | Jane       | 2            | Cleo    | Cat     |

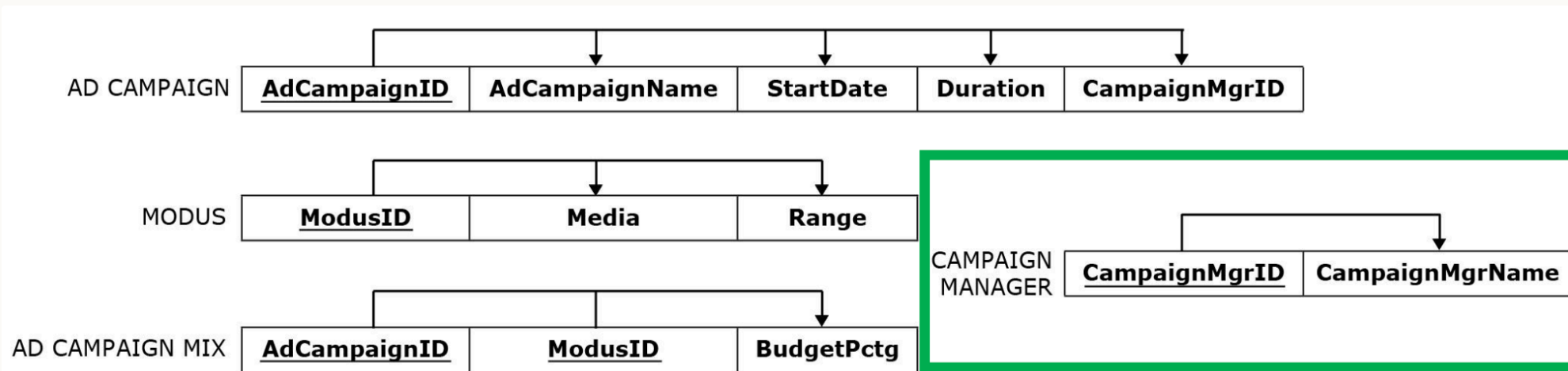
# Normalisation - 2NF

- Second Normal Form (2NF): a table is in 2NF if it is in 1NF and contains no **partial** functional dependencies – one way to ensure this is a single-column primary key
- Normalise by splitting out things that depend on part of the key into a separate table



# Normalisation - 3NF

- Third Normal Form (3NF): A table is in 3NF if it is in 2NF and if it does not contain transitive functional dependencies
- Normalise by splitting out things that depend on nonkey columns into a separate table



Result on the next slide.

### AD CAMPAIGN

| <u>AdCampaignID</u> | AdCampaignName | StartDate | Duration | CampaignMgrID |
|---------------------|----------------|-----------|----------|---------------|
| 111                 | SummerFun20    | 6.6.2020  | 12 days  | CM100         |
| 222                 | SummerZing20   | 6.8.2020  | 45 days  | CM101         |
| 333                 | FallBall20     | 6.9.2020  | 12 days  | CM102         |
| 555                 | AutmnColors20  | 6.9.2020  | 3 days   | CM100         |

### CAMPAIGN MANAGER

| <u>CampaignMgrID</u> | CampaignMgrName |
|----------------------|-----------------|
| CM100                | Roberta         |
| CM101                | Sue             |
| CM102                | John            |
| CM103                | Nancy           |

**Modification Anomaly Resolved:**  
Only one record modified.

### MODUS

| <u>ModusID</u> | Media    | Range    |
|----------------|----------|----------|
| 1              | TV       | Local    |
| 2              | TV       | National |
| 3              | Radio    | Local    |
| 4              | Radio    | National |
| 5              | Print    | Local    |
| 6              | Print    | National |
| 7              | Internet | National |

**Deletion Anomaly Resolved:**  
Campaign 444 deleted but all the data about the campaign manager CM103 and the campaign modus 6 remain in the database.

**Insertion Anomaly Resolved:**  
New campaign modus 7 inserted without inserting an actual campaign using the new modus 7.

### AD CAMPAIGN MIX

| <u>AdCampaignID</u> | <u>ModusID</u> | BudgetPctg |
|---------------------|----------------|------------|
| 111                 | 1              | 50%        |
| 111                 | 2              | 50%        |
| 222                 | 1              | 60%        |
| 222                 | 3              | 30%        |
| 222                 | 5              | 10%        |
| 333                 | 3              | 80%        |
| 333                 | 4              | 20%        |
| 555                 | 3              | 100%       |

# Normalisation Summary

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- Having too many things (columns) in a table can cause anomalies when adding/deleting/changing records
- This occurs when there are partial or transitive functional dependencies
- Solution is to normalise to 3NF:
  1. Identify functional dependencies
  2. Split out columns to remove partial and transitive functional dependencies

*This is probably the trickiest database concept we cover!*

Good news: typically a database that results from a design process (identifying entities, attributes, relationships) is in 3NF

# Today

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1. Data modelling using ER diagrams
2. Translating diagrams to relational databases, and the role of *keys*
3. Anomalies, and avoiding them by *normalisation*

*Next: activities in part 2 today – learn by doing.*

*Next week, building, modifying and querying relational databases with SQL ("es-queue-el", or "sequel").*

# Activities

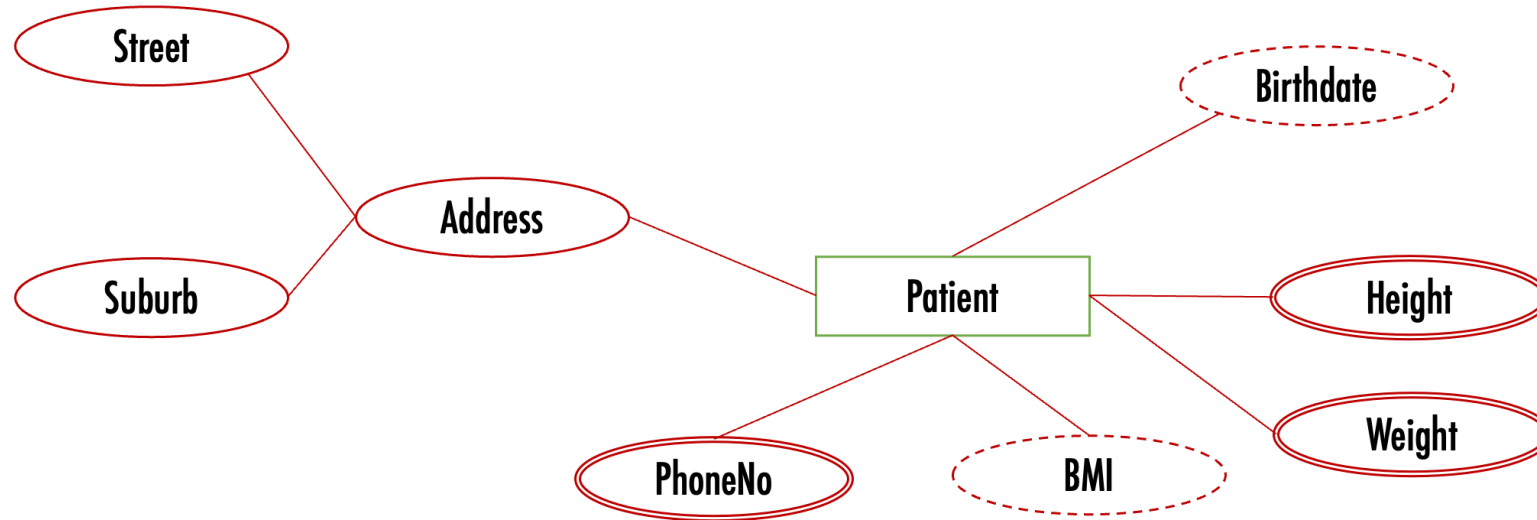
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# Activity – interpret an ER diagram

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*~ 10 minutes, in groups*

- What does the following ER diagram say?
- How might you make it more accurate?





# Activity – sketch an ER diagram

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*~ 15 minutes, in groups*

- **Entities:** Student, Degree, Course, University
- **Relationships:** EnrolledIn, OffersDegree, GraduatedWith

Sketch the ER diagram. What are the **degree** and the **cardinality** of each relationship?

# Activity – sketch it in UML notation

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*~ 10 minutes, in groups*

Sketch a UML diagram showing the same information as the ER diagram you prepared earlier.

How would this ER diagram be mapped to tables in a relational DB?

- **Entities:** Student, Degree, Course, University
- **Relationships:** EnrolledIn, OffersDegree, GraduatedWith

# Activity – normalisation

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*~ 15 minutes, in groups*

- Identify the dependencies in the relation on the next slide – assume the primary key is Flight Number and Day.
- How would you transform it into 3NF? Identify functional dependencies, classify them, and split to remove partial and transitive dependencies.

Assumptions: a flight number always flies at the same time; a flight number does not always use the same plane model. What else do you need to assume?

| Flight Number | Airline      | Day         | Time    | Pilot ID | Pilot Name | Plane model | Plane manufacturer |
|---------------|--------------|-------------|---------|----------|------------|-------------|--------------------|
| NZ123         | Air NZ       | 10 Aug 2018 | 10am    | PL8715   | Pat Smith  | Airbus310   | AirBus             |
| NZ456         | Air NZ       | 11 Aug 2018 | 11am    | PL8716   | Yi Zhang   | Airbus320   | AirBus             |
| YA789         | Yugoslav Air | 17 Aug 2018 | 10:30am | PL9781   | Jim Belich | Boeing737   | Boeing             |
| NZ123         | Air NZ       | 12 Aug 2018 | 10am    | PL9781   | Jim Belich | Boeing737   | Boeing             |
| NZ124         | Air NZ       | 13 Aug 2018 | 11pm    | PL9781   | Jim Belich | Airbus310   | Airbus             |
| YA890         | Yugoslav Air | 17 Aug 2018 | 4pm     | PL9781   | Jim Belich | Airbus310   | Airbus             |
| ...           | ...          | ...         | ...     | ...      | ...        | ...         | ...                |